

| Question | Answer | Mark | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Additional guidance                                                                           |
|----------|--------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 9        | 2820   | P1   | for start to process to find height of triangle,<br>eg $\tan(40) = \frac{h}{5}$ oe<br><b>or</b> equivalent process to find the height of the triangle<br><b>or</b> start to process to find slant height, eg $\frac{10}{\sin 100} = \frac{x}{\sin 40}$                                                                                                                                                                                                           |                                                                                               |
|          |        | P1   | for complete process to find height of triangle, eg $5 \tan 40$ (= 4.19...)<br><b>or</b> complete process to find the slant height, eg $\frac{10}{\sin 100} \times \sin 40$ (= 6.5...)                                                                                                                                                                                                                                                                           | Accept 4.2                                                                                    |
|          |        | P1   | for start of process to find volume of prism,<br>eg $10 \times 20 \times 12$ (= 2400) or $0.5 \times 10 \times "4.19..." \times 20$ (= 419....)<br>or $\frac{1}{2} \times 10 \times "6.52..." \times \sin 40 \times 20$ (419....)<br><b>or</b> process to find total area of cross section,<br>eg $0.5 \times 10 \times "4.19..." + 10 \times 12$ (= 140.9...)<br>or $\frac{1}{2} \times "6.52..." \times "6.52..." \times \sin 100 + 10 \times 12$ (= 140.9...) | $10 \times 20 \times 12$ may be seen as part of a calculation to find the volume of the prism |
|          |        | P1   | for complete process to find total volume,<br>eg $(0.5 \times 10 \times "4.19..." + 10 \times 12) \times 20$                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
|          |        | A1   | for an answer in the range 2810 to 2820                                                                                                                                                                                                                                                                                                                                                                                                                          | If an answer is given in the range in working and then rounded incorrectly award full marks.  |